

The Lie Derivative of Vector Fields in Local Coordinates

Abstract

The coordinate expression for the Lie derivative of a vector field,

$$L_X Y = (X^k \partial_k Y^j - Y^k \partial_k X^j) \partial_j,$$

is derived from first principles. The proof uses only the pullback definition of the flow, the product rule, the chain rule, and the Taylor expansion of the flow. All intermediate steps are shown explicitly.

1 The Pullback Definition and the Flow

Let M be a smooth n -dimensional manifold with local coordinates $(x^1, \dots, x^n)$. The coordinate vector fields are denoted by

$$\partial_i := \frac{\partial}{\partial x^i}, \quad i = 1, \dots, n,$$

and the Einstein summation convention is employed throughout.

Let $X = X^k \partial_k$ be a smooth vector field on M , and let ϕ_t denote its local flow. The flow satisfies

$$\frac{d}{dt} \phi_t(p) = X_{\phi_t(p)}, \quad \phi_0(p) = p.$$

For a vector field Y , the pullback of Y by the flow ϕ_t is defined pointwise by its action on an arbitrary smooth function f :

$$\boxed{(\phi_t^* Y)(f)(p) := Y_{\phi_t(p)}(f \circ \phi_{-t}).} \quad (1)$$

Geometrically, this operation takes the value of Y at the future point $\phi_t(p)$ and transports it back to the starting point p along the inverse flow.

Definition 1 (Lie derivative of a vector field). *The Lie derivative of Y with respect to X is defined as*

$$\boxed{L_X Y := \left. \frac{d}{dt} \right|_{t=0} \phi_t^* Y.} \quad (2)$$

2 The Pullback of a Function Times a Vector Field

Before differentiating, the action of ϕ_t^* on a product fV must be determined.

Proposition 1. *For a smooth function f and a vector field V ,*

$$\boxed{\phi_t^*(fV) = (f \circ \phi_t) \cdot (\phi_t^*V).}$$

Proof. Let g be an arbitrary smooth function, and fix a point p . Using the definition (1) on the vector field fV ,

$$(\phi_t^*(fV))(g)(p) = (fV)_{\phi_t(p)}(g \circ \phi_{-t}).$$

By the definition of the product of a function and a vector field, $(fV)_{\phi_t(p)} = f(\phi_t(p)) \cdot V_{\phi_t(p)}$. Since vector fields act linearly on functions,

$$(fV)_{\phi_t(p)}(g \circ \phi_{-t}) = f(\phi_t(p)) \cdot V_{\phi_t(p)}(g \circ \phi_{-t}).$$

Again by (1), the term $V_{\phi_t(p)}(g \circ \phi_{-t})$ is precisely $(\phi_t^*V)(g)(p)$. Therefore,

$$(\phi_t^*(fV))(g)(p) = f(\phi_t(p)) \cdot (\phi_t^*V)(g)(p).$$

Since this holds for every smooth function g and every point p ,

$$\boxed{\phi_t^*(fV) = (f \circ \phi_t) \cdot (\phi_t^*V).}$$

□

3 The Lie Derivative of a Smooth Function

Before treating vector fields, the action of L_X on functions is required. For a smooth function f , define

$$L_X f := \left. \frac{d}{dt} \right|_0 f \circ \phi_t.$$

Proposition 2. *For any smooth function f ,*

$$L_X f = X(f).$$

Proof. Fix a point p . By definition,

$$(L_X f)(p) = \left. \frac{d}{dt} \right|_0 f(\phi_t(p)).$$

The chain rule yields

$$\left. \frac{d}{dt} \right|_0 f(\phi_t(p)) = df_p \left(\left. \frac{d}{dt} \right|_0 \phi_t(p) \right).$$

From the flow property, $\left. \frac{d}{dt} \right|_0 \phi_t(p) = X_p$. Consequently,

$$(L_X f)(p) = df_p(X_p) = X_p(f).$$

Since this holds at every point p ,

$$\boxed{L_X f = X(f).}$$

In coordinates, $X(f) = X^k \partial_k f$.

□

4 The Derivation Property (Product Rule)

The following product rule is fundamental. Its proof uses the identity established in the preceding section.

Proposition 3. *For a smooth function f and a vector field V ,*

$$\boxed{L_X(fV) = X(f)V + fL_XV.}$$

Proof. Fix a point p . From the pullback identity proved earlier,

$$\phi_t^*(fV)|_p = f(\phi_t(p)) \cdot \phi_t^*V|_p.$$

The derivative with respect to t at $t = 0$ is computed by the ordinary product rule for the two functions of t :

$$L_X(fV)|_p = \left(\frac{d}{dt} \Big|_0 f(\phi_t(p)) \right) \cdot \phi_0^*V|_p + f(\phi_0(p)) \cdot \left(\frac{d}{dt} \Big|_0 \phi_t^*V|_p \right).$$

Using $\phi_0 = \text{id}$ gives $\phi_0^*V|_p = V_p$ and $f(\phi_0(p)) = f(p)$. The first derivative is, by definition, $(L_X f)(p)$, and the second is $(L_X V)_p$. Therefore,

$$L_X(fV)|_p = (L_X f)(p) V_p + f(p) (L_X V)_p.$$

Replacing $(L_X f)(p)$ with $X(f)(p)$ using the preceding proposition yields

$$\boxed{L_X(fV) = X(f)V + fL_XV.}$$

□

5 Computation of $L_X \partial_i$ for a Coordinate Basis Vector

The computation of the Lie derivative of a coordinate vector field ∂_i introduces the characteristic minus sign.

Proposition 4.

$$\boxed{L_X \partial_i = -\partial_i X^j \partial_j.}$$

Proof. Apply $L_X \partial_i$ to an arbitrary smooth function f at a point x in the coordinate chart. Using (1) with $Y = \partial_i$,

$$(L_X \partial_i)(f)(x) = \frac{d}{dt} \Big|_0 \partial_i|_{\phi_t(x)}(f \circ \phi_{-t}).$$

By the chain rule,

$$\partial_i|_{\phi_t(x)}(f \circ \phi_{-t}) = \frac{\partial f}{\partial x^j}(x) \cdot \frac{\partial \phi_{-t}^j}{\partial x^i}(\phi_t(x)).$$

Since $\frac{\partial f}{\partial x^j}(x)$ is independent of t ,

$$(L_X \partial_i)(f)(x) = \frac{\partial f}{\partial x^j}(x) \cdot \left(\frac{d}{dt} \Big|_0 \frac{\partial \phi_{-t}^j}{\partial x^i}(\phi_t(x)) \right). \quad (3)$$

It remains to evaluate the derivative inside the parentheses. The Taylor expansion of the flow about $t = 0$ is

$$\phi_t(x) = x + t X(x) + O(t^2). \quad (4)$$

Differentiating (4) with respect to the spatial coordinate x^i gives the Jacobian of the forward flow:

$$\frac{\partial \phi_t^j}{\partial x^i}(x) = \delta_i^j + t \frac{\partial X^j}{\partial x^i}(x) + O(t^2).$$

Let $J_t(x)$ be the matrix with entries $\frac{\partial \phi_t^j}{\partial x^i}(x)$. Then

$$J_t(x) = I + t B(x) + O(t^2),$$

where I is the identity matrix and $B(x)$ is the matrix $\left[\frac{\partial X^j}{\partial x^i}(x) \right]$.

Since ϕ_{-t} is the inverse of ϕ_t , the chain rule gives

$$J_{-t}(\phi_t(x)) \cdot J_t(x) = I.$$

Thus,

$$J_{-t}(\phi_t(x)) = [J_t(x)]^{-1}.$$

The inverse of $I + tB + O(t^2)$ is $I - tB + O(t^2)$ (by the geometric series expansion). Therefore,

$$J_{-t}(\phi_t(x)) = I - t B(x) + O(t^2).$$

In component form, this is

$$\boxed{\frac{\partial \phi_{-t}^j}{\partial x^i}(\phi_t(x)) = \delta_i^j - t \frac{\partial X^j}{\partial x^i}(x) + O(t^2).} \quad (5)$$

Differentiating (5) with respect to t at $t = 0$ gives the required derivative:

$$\boxed{\frac{d}{dt} \Big|_0 \frac{\partial \phi_{-t}^j}{\partial x^i}(\phi_t(x)) = - \frac{\partial X^j}{\partial x^i}(x).} \quad (6)$$

Substituting (6) into (3) gives

$$(L_X \partial_i)(f)(x) = - \frac{\partial X^j}{\partial x^i}(x) \partial_j f(x).$$

Because this identity holds for every smooth function f and every point x ,

$$\boxed{L_X \partial_i = - \partial_i X^j \partial_j.}$$

□

6 The General Vector Field

Let $Y = Y^j \partial_j$ be an arbitrary smooth vector field. Applying the derivation property from Section 4 to the product $Y^j \partial_j$ gives

$$L_X Y = L_X (Y^j \partial_j) = X(Y^j) \partial_j + Y^j L_X \partial_j.$$

Using $X(Y^j) = X^k \partial_k Y^j$ and $L_X \partial_j = -\partial_j X^k \partial_k$ from the preceding section:

$$L_X Y = X^k \partial_k Y^j \partial_j - Y^j \partial_j X^k \partial_k.$$

In the second term, the dummy indices j and k are renamed: replacing j with k and k with j yields

$$-Y^j \partial_j X^k \partial_k = -Y^k \partial_k X^j \partial_j.$$

Therefore,

$$L_X Y = X^k \partial_k Y^j \partial_j - Y^k \partial_k X^j \partial_j.$$

Factoring out the basis vector ∂_j :

$$\boxed{L_X Y = (X^k \partial_k Y^j - Y^k \partial_k X^j) \partial_j.} \tag{7}$$

The right-hand side of (7) is exactly the coordinate expression for the Lie bracket $[X, Y]$. Hence,

$$\boxed{L_X Y = [X, Y].}$$

7 Origin of the Minus Sign

The minus sign in $L_X \partial_i = -\partial_i X^j \partial_j$ arises from the Taylor expansion of the inverse flow:

$$\phi_{-t}(x) = x - tX(x) + O(t^2).$$

The spatial derivative of the inverse flow is therefore

$$\frac{\partial \phi_{-t}^j}{\partial x^i}(x) = \delta_i^j - t \frac{\partial X^j}{\partial x^i}(x) + O(t^2),$$

which contributes the negative term when differentiated at $t = 0$. This correction term accounts for the change of the coordinate basis itself under the flow.